An Adaptive Framework for Violation Detection Using Multimodal Large Models

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Abstract

Video-based violation detection is critical for industrial safety monitoring, yet prevailing methods rely heavily on object detection models requiring extensive labeled data, leading to high costs and poor generalization. To address these limitations, this paper presents a novel zero-shot detection framework that integrates multimodal large language models (MLLMs) with Retrieval-Augmented Generation (RAG). This framework can **adaptively identify domains and dynamically invoke rules**, demonstrating excellent scalability. Our approach first employs a visual language model to generate standardized semantic descriptions of video content. Subsequently, a multi-granularity semantic retrieval mechanism aligns the real-time video stream with a structured knowledge base of safety regulations. Finally, a dynamic prompt optimization strategy activates the causal reasoning capability of the large model for interpretable violation identification. Experiments on the Action-Tiny dataset demonstrate that our training-free method outperforms the zero-shot MLLM baseline, not only achieves high accuracy but also offers exceptional practicality with easy rule updates and rapid adaptation to new environments, presenting a viable path toward more flexible and efficient intelligent safety monitoring.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Search methodologies..

Keywords

Smart Video Monitoring, Multimodal Large Model, Behavior Analysis, Retrieval-Augmented-Generation (RAG), Production Safety

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AIFM 2025, Guangzhou, China
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ACM ISBN 979-8-4007-1505-1/2025/11
https://doi.org/10.1145/3786709.3786713

ACM Reference Format:
Yuhan Ma, Fangye Wang, Feiyan Yin, Xinyue Li, Zhongguo Yang, and Liping Zhu. 2025. An Adaptive Framework for Violation Detection Using Multimodal Large Models. In *2025 International Conference on Artificial Intelligence and Foundation Model (AIFM 2025), November 14–16, 2025, Guangzhou, China.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3786709.3786713

1 INTRODUCTION

Current video-based violation detection methods have significant shortcomings: rule-based approaches lack semantic understanding capabilities, while deep learning models reliant on annotated data suffer from poor generalization, difficulty adapting to regulatory changes, and high costs [1][2].

The root cause lies in the highly context-dependent nature of safety rules: they vary spatially depending on the scenario, become updated over time with evolving regulations, and become obsolete technologically due to equipment iterations, leading to the rapid obsolescence of static rule sets [3][4].

Notably, the recent emergence of Large Language Models (LLMs) offers new possibilities. With tools like Retrieval-Augmented Generation (RAG) [5]and Knowledge Graphs (KG) [6], we can feed them the latest safety rules — even if those rules change weekly. However, LLMs' effectiveness in specialized domains is hampered by knowledge gaps and static training data, leading to limitations in accuracy and an inability to learn new rules without retraining [7][8].

To address these challenges, this study proposes a **Multimodal Large Model-based Scene-Aware Violation Detection Framework**, whose core innovation is the ability to automatically identify operational contexts and dynamically match them with corresponding regulations, achieved through a three-tier process: first, a multimodal model analyzes video content to generate structured scene descriptions; which is then matched against a dynamic knowledge base by a natural language processing model (e.g., BERT-base-Chinese) to retrieve applicable rules; finally, the system fuses visual evidence with the regulations to produce an interpretable violation judgment. (as illustrated in Figure 1).

This lets one system work across factories, power plants, and construction sites without retraining.

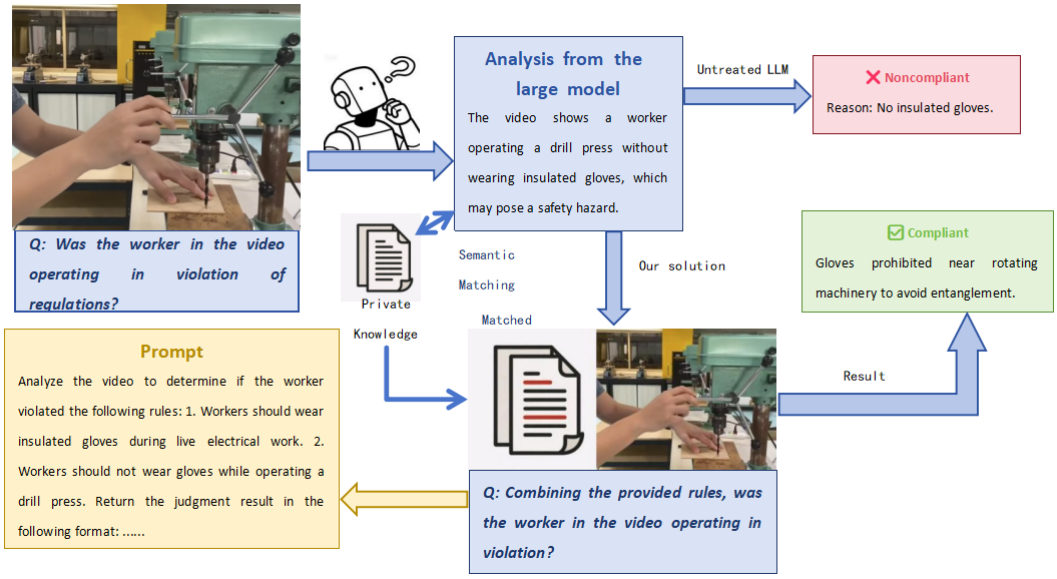

Figure 1: Comparison between our proposed framework and conventional large model approaches.

Our principal **contributions** are:

1. We pioneer the integration of Retrieval-Augmented Generation (RAG) with autonomous domain identification. By proposing a novel multimodal fusion mechanism, our framework can adaptively recognize the operational context and precisely match it with the corresponding safety regulations; 2. We develop an automated knowledge base construction algorithm that generates standardized safety rule bases from minimal natural language input, enabling low-cost system maintenance;3. Comprehensive evaluation on the Action-Tiny dataset demonstrates our framework's superior accuracy and adaptability over conventional methods.

This work proposes an innovative paradigm for dynamic, interpretable, next-generation safety management.

2 RELATED WORK

2.1 Object Detection for Industrial Safety

Early approaches to video-based safety monitoring heavily relied on object detection. The field evolved from traditional feature-based methods like Histograms of Oriented Gradients (HOG) for human detection [9] to deep learning-based models like Faster R-CNN [10] and YOLO [11]. While these detectors excel at locating objects, they struggle to encode complex safety rules (e.g., "must wear insulated gloves") and suffer from performance degradation due to domain shifts across different industrial environments [19]. This lack of semantic understanding and generalization limits their practical applicability, often necessitating costly data relabeling and retraining for each new scenario.

2.2 Multimodal Video Understanding

The advent of Multimodal Large Language Models (MLLMs) has significantly advanced video understanding. Foundational work includes ViViT, which adapted Transformer architectures for video classification [12], and VideoBERT, which pioneered joint modeling of video and language representations [13]. Subsequent works leveraged natural language supervision to learn transferable visual models [14] and employed memory networks with attention mechanisms for video captioning [15]. This paradigm of converting video to text was further extended by incorporating audio modalities [16] and modeling video sequences directly with LLMs [17], proving effective in domains with fixed rules like violence detection [18].

However, these general-purpose MLLMs lack inherent knowledge of specialized safety regulations. Moreover, as highlighted in studies on industrial unsafe behavior [19] and transfer learning surveys [20], existing methods cannot dynamically integrate and reason over textual rules for reliable, zero-shot compliance checking in diverse industrial settings.

3 METHODOLOGY

The workflow of the proposed method is illustrated in Figure 2. This system employs a multimodal large model as the backend server and a multimodal collaborative processing framework as its core architecture. It enables the joint processing of video data, guiding prompts, and textual information from multiple domain-specific scene knowledge bases, thereby improving processing efficiency and adaptability. The framework achieves joint optimization of cross-modal semantic alignment and spatiotemporal feature fusion, offering a comprehensive advantage over traditional methods.

The framework implements dynamic decision-making through a three-tier analysis process:

1. **Spatiotemporal Feature Extraction & Dynamic Domain Identification**: The multimodal large model parses the input video to generate a structured scene description and automatically select the most relevant domain-specific knowledge base.

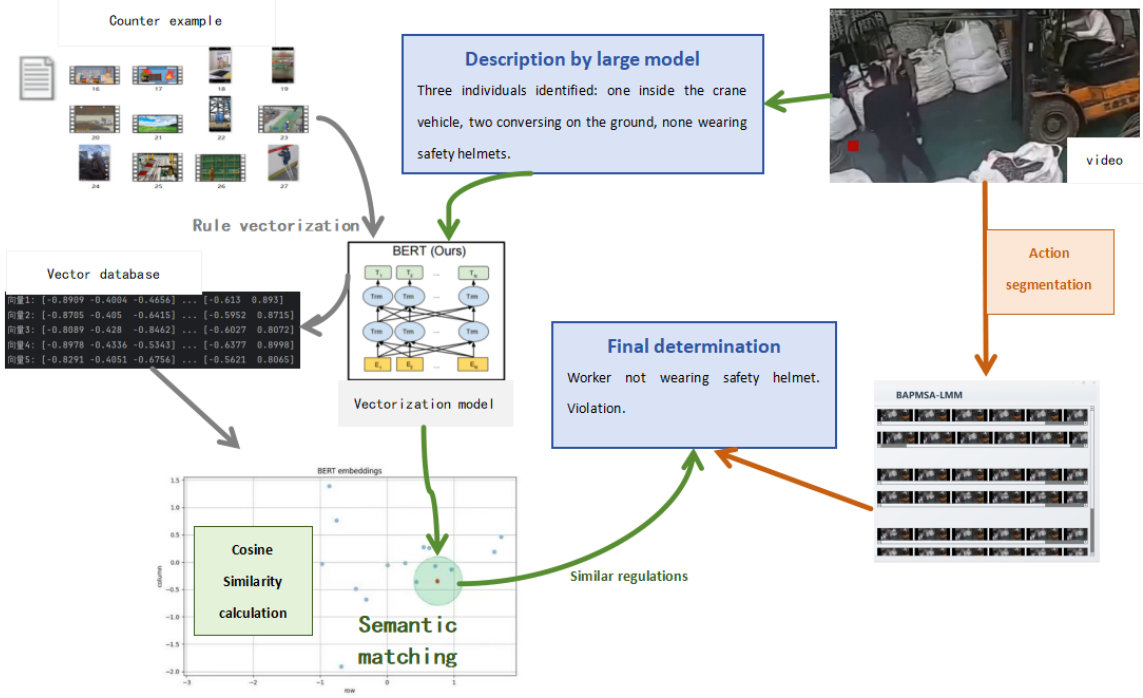

Figure 2: Flowchart of the violation behavior recognition method in videos based on multimodal large models.

2. **Safety Regulation Retrieval**: A natural language processing model (e.g., BERT-BASE-CHINESE) then matches the scene description with a pre-established safety regulation knowledge base, retrieving the most relevant safety requirements.

3. **Action Segmentation & Interpretable Decision-Making**: The system combines the retrieved regulations with the segmented actions to formulate context-aware queries, ultimately generating interpretable judgment criteria.

3.1 Automated Construction of Domain Knowledge Base

To overcome the high cost and scalability limitations of manual knowledge base construction, we propose an automated algorithm that generates a structurally unified safety regulation knowledge base from a small number of video examples and text descriptions. The input to the algorithm is a set of violation cases

$$D = \{(V_i, T_i)\}_{i=1}^{N} \tag{1}$$

Where Vi is a video and Ti is the corresponding textual material. The output is a rule vector library K={(r_j,e_j)}, where r_j is the rule text and e_j is its vector representation.

The core of the algorithm is a three-stage pipeline:

Multi-Factor Scene Parsing: The multimodal large model is employed to parse each video Vi, generating a structured scene description d_i. This description extracts key semantic factors, defined as a tuple

$$d_i = (\text{loc}, \text{person}, \text{action}, \text{equip}, \text{env}, \ldots) \tag{2}$$

Default Condition Induction: The frequency distribution of each factor across all descriptions {di} is calculated. For each factor f, its most frequent value

$$V_f = \text{argmax}_v \text{Freq}\,(v_f) \tag{3}$$

is defined as the default condition for that domain.

Description Compression: Subsequently, each d_i is compressed by removing all factors that match the default condition v_f, retaining only the discriminative features that deviate from the default, resulting in an abbreviated description d_i.

Taking the location factor as an example, the workflow of the above approach is illustrated in Figure 3.

Finally, the pooler_output of the bert-base-chinese model is leveraged to obtain sentence vector representations for all documents, constructing the semantic space of the knowledge base.

3.2 Core Workflow

Our method adopts a three-level cascaded analysis architecture, taking a video stream V as input and ultimately outputting a set of violation detection results R.

3.2.1 Multimodal Feature Parsing Layer. Given an input video V, the multimodal large model's visual encoder first extracts spatiotemporal features from key frames. Let

$$F = \{f_t\}_{t=1}^{T} \tag{4}$$

be the frame-level feature sequence. By employing temporal modeling (e.g., a Transformer encoder) to capture motion context, the

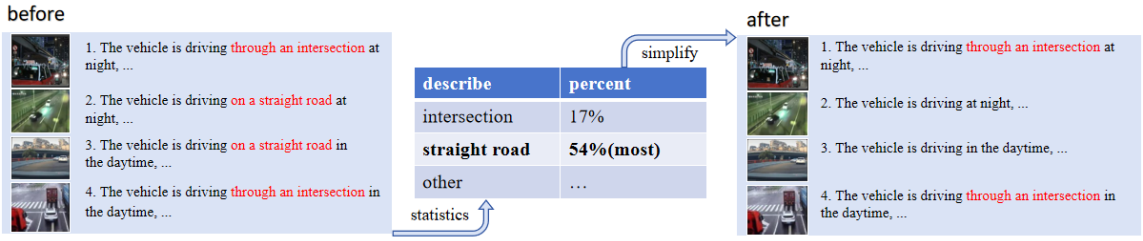

Figure 3: Conversion process from violation videos to safety regulation text (using site factors as an example).

model outputs a structured scene description S:

$$S = MM - LLM\,(V) \tag{5}$$

Where S contains the set of detected persons P={p_k} and their associated semantic information (actions, interacting objects, etc.).

Concurrently, the model is prompted to generate a high-level domain summary and extract core keywords KW_video from the initial parsing results. The system then computes the similarity between KW_video and the keyword sets of all available knowledge bases. The knowledge base with the highest similarity is automatically selected for subsequent processing (the target domain can also be directly specified by the user).

This integrated process ensures that all downstream analysis is grounded in the correct contextual domain from the outset.

3.2.2 Regulatory Knowledge Matching Layer. The system utilizes a natural language processing model (e.g., BERT-BASE-CHINESE) to encode the descriptive text into 768-dimensional vectors. Subsequently, it calculates the cosine similarity between the query vector and each vector in the pre-stored knowledge base matrix:

$$\text{similarity} = \frac{\sum_{i=1}^{768} A_i * B_i}{\sqrt{\sum_{i=1}^{768} (A_i)^2} * \sqrt{\sum_{i=1}^{768} (B_i)^2}} \tag{6}$$

For each personnel description vector B, the system retrieves the top_k (k=3) most similar clauses A from the knowledge base matrix for subsequent judgment.

3.2.3 Decision Generation Layer. In the Decision Generation Layer, the system constructs an enhanced prompt using the matched clauses as context. This prompt, along with the original video, is fed to the large model with explicit instructions to judge violations and provide reasoning.

The entire process achieves fine-grained analysis through a two-stage prompt engineering approach. The first stage focuses on accurate description, while the second stage combines semantic matching results for professional judgment. This ensures both the accuracy of video understanding and the proper application of safety regulations.

4 EXPERIMENT

The remarkable capabilities of Multimodal Large Language Models (MLLMs) in video understanding tasks provide a solid foundation for building knowledge-driven intelligent video analysis solutions [21][22]. Particularly in domains like industrial safety monitoring, where regulations are well-defined but scenarios are highly variable, the key to improving violation recognition accuracy lies in the deep integration of the general knowledge of MLLMs with domain-specific rules. Recent studies have shown that instruction fine-tuning [23] or phased reasoning [24] can effectively guide MLLMs to handle specific tasks, while the successful application of Retrieval-Augmented Generation (RAG) in specialized fields like healthcare and law [25][26] validates the effectiveness of external knowledge injection for enhancing model judgment. Based on these insights, we propose an adaptive violation recognition scheme that integrates visual analysis, knowledge base retrieval, and two-stage reasoning.

4.1 Experiments Setting

Dataset: Experiments were conducted on the public industrial violation video dataset, Action-Tiny, which contains 100 clips for each of 11 categories. We selected 900 videos across 4 representative violation recognition tasks for validation, including perimeter intrusion, safety belt compliance, smoking ban enforcement, and unsupervised ladder use.

Evaluation Metrics: Accuracy, Precision, Recall, and F1-score were adopted to comprehensively evaluate model performance.

Implementation Details: Our scheme begins with event segmentation of the input video (24 FPS for segmentation, 4 FPS for recognition). We employed Qwen-VL-Max as the MLLM for scene recognition and reasoning, and used the bert-base-chinese model to encode both scene descriptions and regulatory clauses into semantic vectors for building the knowledge base.Implementation details are as follows:

MLLM Zero-shot Baseline: Video frames were directly fed into the Qwen-VL-Max model for zero-shot violation judgment without introducing any domain knowledge or specialized reasoning mechanisms.

4.2 Results and Analysis

The performance comparison of different methods on the four tasks is presented in Table 1.

Analysis: The **MLLM Zero-shot** Baseline demonstrates strong generalization ability, achieving high recall across multiple tasks, which indicates its great potential in detecting potential violations. However, its precision remains consistently low, leading to numerous false positives, suggesting that MLLMs lacking domain knowledge cannot accurately discern the subtle differences between compliant and violating behaviors.

Table 1: Performance comparison of different methods on the Action-Tiny dataset (%).

Detection test	Method	Accuracy	Precision	Recall	F1-score
safety-belt	MLLM	70.7	53.5	91	67.4
	Our Method	**88.3**	**76**	**95**	**84.4**
cross_fence	MLLM	70	72.7	64	68
	Our Method	**93**	**95.7**	**90**	**92.8**
smoking	MLLM	37	38.2	42	40
	Our Method	**68.5**	**70.3**	**64**	**67**
using ladder	MLLM	68.5	68	70	68.9
	Our Method	**86.5**	**84.8**	**89**	**86.8**

Table 2: Performance comparison of different model configurations (%).

#	Model Configuration	Accuracy	Precision	Recall	F1-score
1	Baseline	63.2	55.9	70.8	62.4
2	#1+Dynamic Domain Identification	73.3	69.7	75.6	72.5
3	#2+Automated Knowledge Base	78.3	76.5	78.9	77.7
4	#3+Two-Stage Prompting (Full)	85.1	79.6	86.6	82.9

Our Proposed Method achieves the best overall performance (highest F1-score) across all tasks. It retains the high recall advantage of MLLMs while significantly improving precision through the incorporation of a domain knowledge base and a two-stage reasoning mechanism, fully validating the effectiveness of knowledge enhancement and structured reasoning for improving the accuracy of industrial violation recognition.

In summary, our solution successfully integrates the stability of visual models with the deep reasoning capabilities of large models, refining performance through domain knowledge injection. It significantly improves precision while maintaining a high recall rate, making it more suitable for industrial safety supervision scenarios with stringent accuracy requirements.

4.3 Ablation Studies

The ablation studies in this section are based on the dataset described in Section 4.2, comprising four categories and 900 video instances. The reported results are derived from weighted averages, aiming to validate the effectiveness of each core module. The results are in Table 2.

Our full model achieves major gains across all metrics, with an accuracy of 85.1% and a balanced F1-score of 82.9%. Dynamic Domain Identification is the primary driver, drastically improving precision from 55.9% to 69.7%. The Automated Knowledge Base further refines all metrics steadily. Finally, Two-Stage Prompting delivers a remarkable boost in recall to 86.6%, cementing the overall performance.

5 CONCLUSION

In this paper, we address the challenge of applying general-purpose Multimodal Large Language Models (MLLMs) to the dynamic and context-sensitive domain of industrial safety monitoring. By introducing a novel framework that synergizes semantic video understanding with a Retrieval-Augmented Generation (RAG) mechanism, we achieve accurate, training-free violation detection. Specifically, through the automated construction of a structured knowledge base and a cascaded reasoning process that dynamically retrieves and integrates relevant safety regulations, our method enables MLLMs to perform interpretable, knowledge-grounded judgments. Experimental results on the Action-Tiny dataset demonstrate that our approach significantly enhances precision over zero-shot MLLM baselines while maintaining high recall, effectively balancing generalization capability with regulatory accuracy. In the future, we will explore scaling the knowledge base to cover more complex scenarios and optimizing the retrieval efficiency for real-time deployment.

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